

Validating a Proprietary Sepsis Prediction Model: Lessons From External Evaluation

Student Name: Aasish Katwal

Indiana Wesleyan University

5.3 Assignment: Model Validation and Verification

Validating a Proprietary Sepsis Prediction Model: Lessons From External Evaluation

Case and Data Preparation

Wong et al. (2021) independently evaluated the Epic Sepsis Model (ESM), a penalized logistic-regression model trained by its vendor on 405,000 encounters from three health systems (2013–2015). The external cohort comprised 27,697 adults and 38,455 Michigan Medicine hospitalizations from December 2018 to October 2019; sepsis occurred in 6.6%. This is an especially useful published case because the model was already embedded in hundreds of hospitals, yet its proprietary design limited scrutiny.

Preprocessing began with a defensible cohort and target. Investigators included adult admissions, excluded units where alerts were active to avoid intervention-induced bias, and constructed a composite sepsis label from CDC surveillance criteria or ICD-10 sepsis plus temporally aligned inflammatory and organ-dysfunction criteria. Scores were generated every 15 minutes; post-onset scores were removed from the primary analysis to prevent target leakage. The publication does not disclose the vendor’s feature cleaning, missing-data treatment, scaling, or variable encoding. That opacity is itself a verification failure: reproducible preprocessing and a data dictionary should be contractual deployment requirements.

Model Choice and Validation Design

Penalized logistic regression is a reasonable choice for high-frequency EHR prediction because regularization limits overfitting and coefficients can be more auditable than those of a complex ensemble. Its simplicity, however, cannot correct label ambiguity, temporal drift, site

differences, or poorly specified inputs. A transparent local baseline should therefore be compared with ESM before deployment.

The study performed temporal and geographic external validation rather than relying on the developer's internal split. It evaluated the complete hospitalization trajectory and 4-, 8-, 12-, and 24-hour horizons; tested thresholds; inspected calibration; estimated lead time; compared alerts with timely antibiotics as usual care; quantified alert fatigue; and ran a leakage-sensitive analysis that admitted scores up to three hours after onset. This design verifies clinical usefulness as well as statistical fit. A remaining weakness is single-center validation and a debated composite reference standard; multicenter prospective silent trials and bootstrap confidence intervals for subgroup metrics should precede broad use.

Performance, Trade-offs, and Interpretation

At the operational threshold of 6, hospitalization-level AUC was 0.63 (95% CI [0.62, 0.64]), sensitivity 33%, specificity 83%, positive predictive value (PPV) 12%, and negative predictive value 95%; calibration was poor (Wong et al., 2021). Because prevalence was low, the high negative predictive value is not reassuring by itself. Raising the threshold would reduce false alarms and clinician burden but miss more sepsis; lowering it would improve recall while worsening PPV and alert fatigue. At threshold 6, 18% of admissions would alert, eight patients required evaluation per true case, 67% of sepsis cases were missed, and only 7% of cases lacking timely antibiotics were newly identified. The AUC rose to 0.80 when post-onset scores were allowed, demonstrating how leakage can create a misleading performance claim. Decision-curve net benefit, workload per true positive, calibration, and time-to-action should therefore accompany AUC.

Ethical Safeguards and Post-deployment Monitoring

A safe implementation would minimize EHR fields to the intended use, encrypt data in transit and at rest, restrict role-based access, log queries, de-identify evaluation extracts, and prohibit secondary use without governance. Fairness testing should report sensitivity, specificity, PPV, false-negative rate, calibration, and alert burden by race/ethnicity, sex, age, disability, language, insurance status, and care site, with uncertainty intervals. Disparities require root-cause review, threshold or workflow changes, and—if harms cannot be reduced—suspension. Clinicians must retain override authority, receive uncertainty and limitation disclosures, and never treat an alert as a diagnosis.

Deployment should begin in silent mode with predefined acceptance gates. A multidisciplinary owner should review weekly dashboards for missingness, range violations, input drift, alert volume, latency, overrides, antibiotic timing, and adverse events; monthly reviews should recompute discrimination, calibration, subgroup error gaps, PPV, sensitivity, and net benefit. Trigger investigation for material drift or a sustained breach of locally approved safety/fairness bounds; freeze updates, revert to the prior version, or disable alerts when patient safety is threatened. Quarterly revalidation, annual independent audit, versioned model cards, incident reporting, and controlled recalibration align with continuous lifecycle monitoring recommended by NIST (Tabassi, 2023).

Critical Alignment and Practical Lessons

The case aligns with course best practices by separating development from external testing, preventing temporal leakage, reporting confidence intervals and calibration, testing thresholds, and connecting metrics to workflow. It conflicts with those practices through proprietary preprocessing, limited reproducibility, reliance on one external site, weak subgroup

reporting, and adoption before independent validation. The central lesson is that integration convenience is not evidence of validity. Organizations should define the intended use and harm-weighted acceptance criteria first, validate locally and prospectively, compare against usual care, document every data transformation, and monitor the human–AI system—not merely the algorithm.

References

- Tabassi, E. (2023). Artificial intelligence risk management framework (AI RMF 1.0) (NIST AI 100-1). National Institute of Standards and Technology.
<https://doi.org/10.6028/NIST.AI.100-1>
- Wong, A., Otles, E., Donnelly, J. P., Krumm, A., McCullough, J., DeTroyer-Cooley, O., Pesttrue, J., Phillips, M., Konye, J., Penozza, C., Ghous, M., & Singh, K. (2021). External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. *JAMA Internal Medicine*, 181(8), 1065–1070.
<https://doi.org/10.1001/jamainternmed.2021.2626>